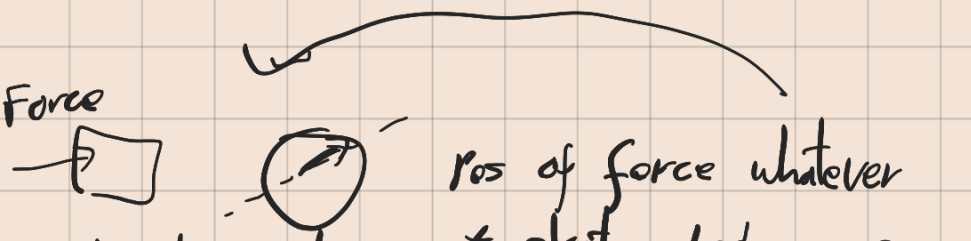
Kynematics

Rigid body

Force



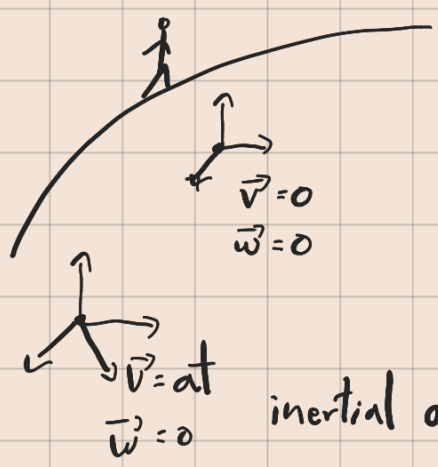
Force shifted on action line

Pos of force whatever

≠ elastic body

1. Assumption

2. Assumption



during change

absolute observation

⟹ fixed on Earth

≙ can apply Newton's Law

inertial observer

$$\vec{r}_3 = \vec{r}_1 + \vec{r}_{12} + \vec{r}_{23} \qquad \vec{v}_3 = \frac{d}{dt}\vec{r}_3 = \frac{d}{dt}\left(\vec{r}_1 + \vec{r}_{12} + \vec{r}_{23}\right)$$

$$= \frac{d}{dt}\vec{r}_1 + \frac{d}{dt}\vec{r}_{12} + \frac{d}{dt}\vec{r}_{23}$$

$$= \vec{v}_1 + \vec{v}_{12} + \vec{v}_{23}$$

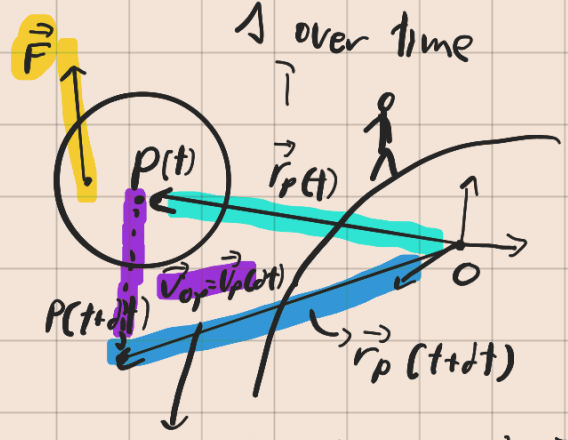
Vectors and Represent. of Vectors

1. Observation

$\vec{r}_{op} = \vec{r}_p$

$\vec{r}_p(t+dt)$

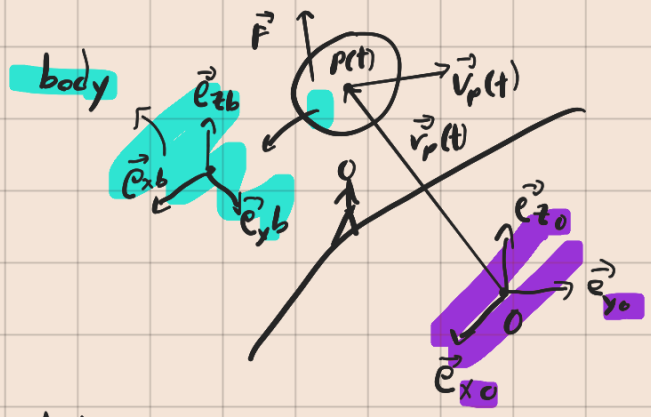
$\vec{F}$



Observed by someone w/o Δ in $\vec{r}$ or $\vec{a}$

$$= \lim_{dt\to 0} \frac{\vec{r}_p(t+dt) - \vec{r}_p(t)}{dt} = \frac{d}{dt}\vec{r}_p$$

Need for coord. systems to describe position



Inertial (zero) coord. sys.
(where observer is)

2. Representation

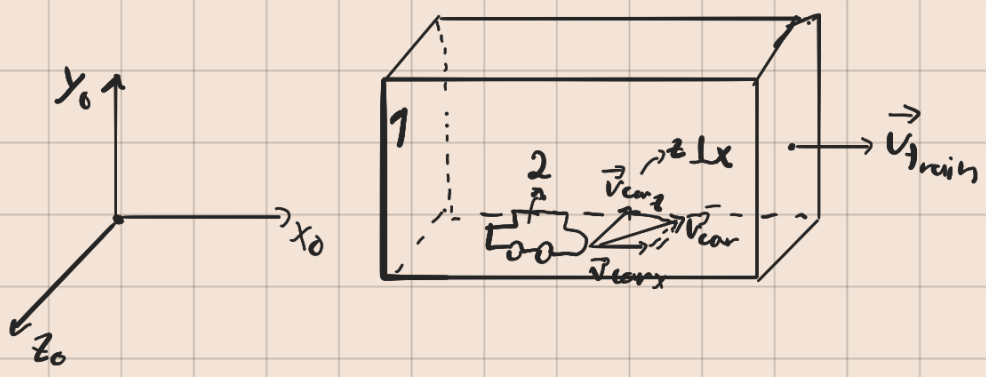
$$\vec{v}_p = x_{v_b}\,\vec{e}_{x_b} + y_{v_b}\,\vec{e}_{y_b} + z_{v_b}\,\vec{e}_{y_c}$$ → vector (geometric)

Coordinates

(no arrow)

$${}_bV_p = \begin{bmatrix} x_{v_b} \\ y_{v_b} \\ z_{v_b} \end{bmatrix}$$ representation of $\vec{v}_p$ in coord. Sys. b

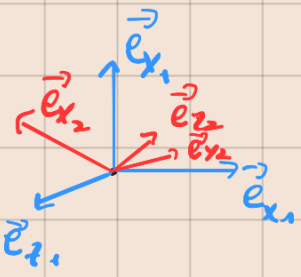
Change of base is possible between coord. sys. → important for derivative



$${}_0V_{12} = \Big[$$

Base Transformation of Vectors

$$\left(\vec{r}_{PQ} = x_1 \vec{e}_{x_1} + y_1 \vec{e}_{y_1} + z_1 \vec{e}_{z_1}\right)$$



$$\hookrightarrow {}_1r_{PQ} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$$

$${}_1r_{PQ} = x_1 \, {}_1e_{x_1} + y_1 \, {}_1e_{y_1} + z_1 \, {}_1e_{z_1}$$

$$= x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + y_1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + z_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$$

$${}_2r_{PQ} = x_1 \cdot {}_2e_{x_1} + y_1 \cdot {}_2e_{y_1} + z_1 \cdot {}_2e_{z_1}$$

transf. Matrix

$$= \begin{bmatrix} \vdots & \vdots & \vdots \\ {}_2e_{x_1} & {}_2e_{y_1} & {}_2e_{z_1} \\ \vdots & \vdots & \vdots \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = {}_2A_1 \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

Properties of Rotation Matrices

Non commutative

$${}_1A_3 = {}_1A_2 \, {}_2A_3$$

$$\neq {}_2A_3 \, {}_1A_2$$

Length invariant

$${}_1r^T \, {}_1r = {}_2r^t \, {}_2r \quad \text{(scalar prod.)}$$

$$= ({}_2A_1 \, {}_1r)^t \, {}_2A_1 r_1$$

$$= {}_1r^t \underbrace{{}_2A_1^t \, {}_2A_1}_{\overset{!}{=} I} r_1$$

$${}_2A_1^t \, {}_2A_1 = I$$

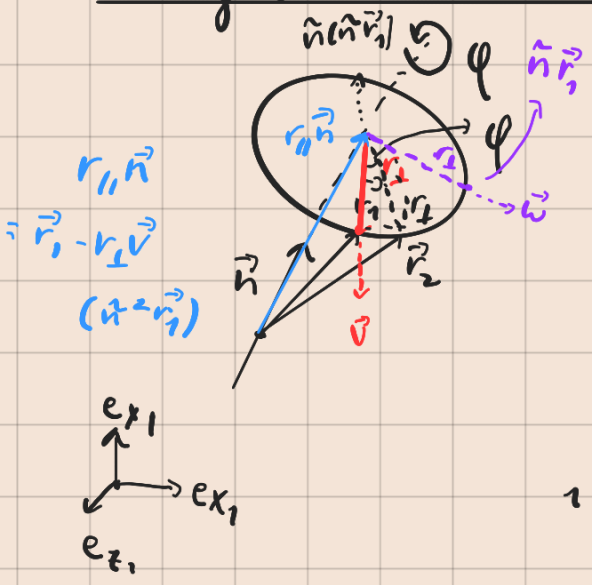
orthonormal matrices

$${}_2A_1^{-1} = {}_2A_1^t = {}_1A_2 \quad \text{(pica)}$$

$$({}_aA_b \, {}_bA_c)^{-1} = {}_bA_c^{-1} \, {}_aA_b^{-1} = {}_bA_c^t \, {}_aA_b^t = ({}_aA_b \, {}_bA_c)^t$$

Euler's Rotation Theorem

Rodrigues' Rotation Formula



cross-prod. $\vec{n} \times \vec{r}_1$

$$\vec{r}_1 = r_{\parallel}\vec{n} + r_{\perp}\vec{v}$$

$$\vec{r}_2 = r_{\parallel}\vec{n} + \cos\varphi\, r_{\perp}\vec{v} + \sin\varphi\, \underbrace{r_{\perp}\vec{\omega}}_{\tilde{n}\vec{r}_1}$$

$$(\mathbb{1}+\tilde{n}^2)\vec{r}_1 - \cos\varphi\, \tilde{n}^2 r_1 + \sin\varphi\, \tilde{n}\vec{r}_1$$

$$= (\mathbb{1} + \sin\varphi\, \tilde{n} + (1-\cos\varphi)\tilde{n}^2)\vec{r}_1$$

$${}_1r_2 = \underbrace{(\mathbb{1} + \sin\varphi\, {}_1\tilde{n} + (1-\cos\varphi)\, {}_1\tilde{n}^2)}_{A \text{ (rotation matrix)}}\, {}_1r_1$$

$${}_1e_{x_2} = A\, {}_1e_{x_1} = A \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

eu odeio você //